Reflexive Descent to Minimal Seeds: Uniform Bounds and Free Sums

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Abstract

We study saturated descents through reflexive lattice subpolytopes to r-minimal seeds. The uniform-existence conjecture has an affirmative answer: every descent terminates, and a published volume bound gives the classification-free estimate $D_d \leq 2(s_d - 1)^2 - 2$ for $d \geq 4$. We give an exact finite construction of a descent and prove that the reflexive-subpolytope order of a free sum in a direct-sum lattice is the product of the factor orders. Shortest and longest descent lengths are therefore additive. An explicit family in dimension $2r$ has shortest length r and longest length $2r$, proving that both depth and the gap between these lengths can grow with dimension. A second base, checked by two exact integer enumerators, gives lengths $5r$ and $6r$ and a complete formula for the descent counts. The extremal depths are superadditive, and D_d/d has an extended limit. In dimension four the earlier bound of 348 for every descent remains valid; twelve interval certificates refine the shortest-depth range to $22 \leq D_4 \leq 346$ relative to the supplied Kreuzer–Skarke census. The exact value of D_4 and the optimal growth rate are not determined.

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The earlier joint manuscript with Ushashi Bhattacharya, *Reflexive Descent to Minimal Seeds in the Kreuzer–Skarke List*, Hexagon 2610.00456v1, supplies the preceding descent framework and finite-dimensional certificates. Her contribution to that earlier work is acknowledged here. This revision adds the free-sum results and explicitly distinguishes the original results from the new checks.

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1. Introduction

A lattice polytope is reflexive when it contains the origin in its interior and every facet lies on a hyperplane $\langle x, y\rangle = -1$ with y a lattice vector; equivalently, its polar dual is again a lattice polytope. Batyrev showed that a reflexive polytope Δ of dimension four gives a family of Calabi–Yau threefolds, the anticanonical hypersurfaces in the toric variety of the fan over the faces of Δ^*, and that the exchange of Δ and Δ^* is mirror symmetry [3]. Kreuzer and Skarke classified these polytopes: there are 16 in dimension two, 4,319 in dimension three [5] and 473,800,776 in dimension four [6], up to unimodular equivalence. We write KS_d for the list in dimension d.

The classification rests on a finiteness principle. A lattice polytope with the origin in its interior is called *minimal* when removing any one of its vertices and taking the convex hull of the others loses that property, and every reflexive polytope contains a minimal one spanned by some of its vertices [4] and [6]. In dimension four Kreuzer and Skarke worked with a sharper notion: a reflexive polytope is *r-minimal* when it contains no reflexive polytope other than itself, and every reflexive polytope contains an r-minimal one. Both notions describe the seeds from which the whole list is grown, and the question studied here is how far a polytope is from its seeds.

The seed universality preprint [17] proposed that the list is generated from at most 64 seed orbits with Hodge numbers constant on each orbit. A later note [18] refuted that proposal, kept what is true near it, and isolated the following statement as the part that remained open.

Conjecture 1 (Uniform transition depth [18]). There is an absolute constant D such that every $\Delta \in \mathrm{KS}_4$ is connected to some minimal seed by a chain of at most D subpolytope steps $\Delta_i \supset \Delta_{i+1}$ in which every intermediate hull is reflexive after unimodular equivalence. The stratification data suggest a value of D of order a few tens.

The assertion of a uniform constant is already settled by the finite-order argument below; it is distinct from determining the sharp constant or justifying the heuristic of a few tens. This revision gives a further, classification-free construction for varying dimensions.

The note also asserted that descent through reflexive polytopes alone is not always possible, and offered a vertex-deletion chain with non-reflexive intermediate hulls as the substitute. We show that the assertion is false, prove the conjecture with an explicit constant, and determine the constant exactly in dimensions two and three.

To make the statement precise we use the order of the reflexive polytopes by inclusion. A reflexive polytope $\Gamma \subsetneq \Delta$ is a *cover* of Δ when no reflexive polytope lies strictly between them.[1] A *reflexive descent* of Δ is a chain

$$\Delta = \Delta_0 \supsetneq \Delta_1 \supsetneq \cdots \supsetneq \Delta_k$$

in which each Δ_{i+1} is a cover of Δ_i and Δ_k is r-minimal. Its length is k, and the *transition depth* $\delta(\Delta)$ is the least length of a reflexive descent of Δ. A descent never skips a reflexive polytope, so it is the strictest form of the chains in Conjecture 1; without the cover condition every polytope reaches a seed in one step, as we explain in Section 3. We write D_d for the largest transition depth in KS_d, and $|P|$ for the number of lattice points of a polytope P. For arbitrary dimension, D_d denotes the maximum over all reflexive d-polytopes; no classified list in dimension greater than four is assumed. Its finiteness also follows from Proposition 15.

Theorem 2. *Every reflexive polytope has a reflexive descent, every reflexive descent ends at an r-minimal polytope, and every r-minimal polytope*

[1] Here and below all polytopes live in the same lattice M and have the origin as their only interior lattice point; a subpolytope of Δ is a lattice polytope contained in Δ. Reflexivity of a subpolytope is always meant with respect to M and the origin, which is the reading of "after unimodular equivalence" in Conjecture 1: a unimodular map that carries Γ into Δ fixes the origin and the lattice.

contains a minimal polytope. Hence every $\Delta \in \mathrm{KS}_4$ *is joined to a minimal seed by a chain in which every intermediate hull is reflexive and no reflexive polytope fits between two consecutive members.*

THEOREM 3. *No reflexive descent in dimension four has more than* 348 *steps. More precisely, every reflexive descent of* $\Delta \in \mathrm{KS}_4$ *has length at most* $\min(|\Delta| - 6,\ 680 - |\Delta^*|)$ *and at most the pair bound* $L(|\Delta|, |\Delta^*|)$ *of Section 4. The first number never exceeds* 654 *and the second never exceeds* 348*, and each maximum is attained by a single polytope. Hence* $D_4 \leq 348$*, and Conjecture 1 holds with* $D = 349$*.*

THEOREM 4. $D_2 = 5$ *and* $D_3 = 11$*. In dimension two the maximum is attained only by the triangle with ten lattice points; in dimension three it is attained by exactly two polytopes, with* 35 *and* 31 *lattice points, both of which are contained in no larger reflexive polytope.*

THEOREM 5. *The polytopes* P_1 *and* P_2 *of Section 7.2, with* 37 *lattice points each, have transition depth exactly* 22*, and* P_1 *is unimodularly equivalent to the polytope reached after ten steps of a reflexive descent of the simplex* Δ_{680} *with* 680 *lattice points, the largest reflexive polytope in dimension four. The transition depth of* Δ_{680} *is at most* 31*. Hence* $22 \leq D_4 \leq 348$*.*

Theorem 3 bounds every descent, not only the shortest one, so it also bounds the longest chain of covers below any polytope. The existence of some constant D is the easy part of the conjecture: the list is finite and Theorem 2 gives each polytope a descent. The content of Theorem 3 is that the constant can be read off from two numbers attached to each polytope, its lattice points and the lattice points of its dual, without examining any polytope. The same count, with Blichfeldt's bound on lattice points and the volume theorem of Balletti, Kasprzyk and Nill in place of the list, bounds the depth in every dimension: $D_d \leq 2(s_d - 1)^2 - 2$ for $d \geq 4$, where s_d is Sylvester's sequence (Proposition 15). The guess of "a few tens" in the conjecture is neither confirmed nor refuted. The exact values 5 and 11 in lower dimension, the depth 22 of two polytopes with thirty-seven points and the descent of length 31 from the largest polytope are consistent with it. The interval refinement in Corollary 14 gives $22 \leq D_4 \leq 346$ relative to the supplied census; the exact value remains open.

Theorem 2 is a short argument in the inclusion order (Section 3). Theorem 3 counts lattice points on both sides of the duality: along a descent the polytope loses points and its dual gains them, and both counts must stay among the pairs that occur in the list (Section 4). For the lower bounds we need all covers of a given polytope. Section 5 describes them as cuts along dual lattice points; the volume bound of Balletti, Kasprzyk and Nill [14] leaves finitely many cuts to try, and a climbing lemma certifies one cover without listing the others. In

dimension two these tools give $D_2 = 5$ by hand, and in dimensions three and four they give the exact values and constructions of Sections 6 and 7. A lower bound for a depth is a statement about every descent, and in dimension four we prove it with a certificate: a finite set of polytopes with their complete lists of covers and a bound attached to each, subject to a condition that can be checked one polytope at a time (Lemma 22). The polytopes P_1 and P_2 are skew bipyramids over a three-dimensional reflexive tetrahedron, and their certificate contains 5,592 polytopes. Section 9 lists what remains open.

THEOREM 6 (Free-sum construction). *For reflexive polytopes P, Q in direct-sum lattices,*

$$R(P \oplus Q) \cong R(P) \times R(Q), \qquad \delta(P \oplus Q) = \delta(P) + \delta(Q).$$

The analogous equality holds for longest descent lengths. In particular, explicit polytopes of dimension $2r$ have shortest and longest lengths r and $2r$; a second independently checked base gives $5r$ and $6r$. The maxima satisfy $D_{m+n} \geq D_m + D_n$.

These statements are proved in Section 8. The first family has a five-element planar subpolytope order, verified directly by primitive facet inequalities, so it needs neither a census nor a computational certificate. It answers the question about the existence of a growing lower bound from the earlier manuscript [1]. The second family's base profile is exhaustively checked over 512 subsets by two integer implementations. These are polytope descent results; they do not construct a string vacuum or settle GSC.

History. Reflexive polytopes entered algebraic geometry through Batyrev's construction of mirror pairs of Calabi–Yau hypersurfaces [3]; the lattice-point formulas for $h^{1,1}$ and $h^{2,1}$ in that paper made the polytopes themselves the objects of study. Kreuzer and Skarke set out to list them. Their first paper [4] introduced the minimal polytopes, showed that a minimal polytope is either a simplex or a union of lower-dimensional simplices sharing the origin as an interior point, and attached a weight system to each simplex, so that the minimal polytopes in a given dimension come from finitely many combined weight systems. The three-dimensional list of 4,319 polytopes followed in 1998 [5]. In dimension four the combined weight systems number 201,346, too many to use directly, and Kreuzer and Skarke replaced minimality by r-minimality: only 308 combined weight systems give r-maximal polytopes, that is, duals of r-minimal ones, and together with 25 sublattice versions and 7 exceptional polytopes they give the 340 r-maximal reflexive polytopes of dimension four [6]. Every reflexive four-polytope is a subpolytope of one of these, and the complete list of 473,800,776 was obtained by enumerating

subpolytopes with the program PALP [7]. The lattice-point counts range from 6 to 680 [6].

The inclusion order was used from the start to argue that the Calabi–Yau hypersurfaces form a connected web. Candelas, Green and Hübsch had shown that the complete intersections in products of projective spaces are connected by conifold transitions [8]; Avram, Kreuzer, Mandelberg and Skarke showed that the toric hypersurfaces from weighted projective spaces form a web connected by inclusions of polytopes [9]; and Kreuzer and Skarke observed that connectedness of all reflexive four-polytopes reduces to connectedness of the 340 r-maximal ones, which they checked during the classification [6]. Reid's question whether all Calabi–Yau threefolds lie in one family connected by such transitions [10] remains open in general. Databases built on the list [11] made the joint statistics of polytopes and Hodge numbers easy to read, and the list is now distributed in a form that can be streamed in a few minutes [12].

The extremes of the list have a separate history. Lagarias and Ziegler bounded the volume of lattice polytopes with a fixed number of interior points [13]; for reflexive polytopes the sharp bound was proved by Balletti, Kasprzyk and Nill [14], who showed that in every dimension $d \geq 4$ the largest reflexive polytope is a simplex built from Sylvester's sequence. In dimension four it is the simplex Δ_{680} with 680 lattice points, the Newton polytope of the hypersurface of degree 84 in the weighted projective space $\mathbb{P}(1,1,12,28,42)$, whose mirror is the Calabi–Yau threefold with the largest value of $h^{1,1}$ in the list.

The question of distance to the seeds came from the seed universality preprint [17]. Its orbit picture combined polar duality, stellar subdivision and flips, and claimed constant Hodge numbers on at most 64 orbits; the correcting note [18] disposed of that claim with two observations (duality exchanges the Hodge pairs $(1,101)$ and $(101,1)$ of the quintic and its mirror, and the list realises 30,108 Hodge pairs), proved the statements that survive at a fixed polytope, and stated Conjecture 1. The present paper answers it. Related work of the first author on Calabi–Yau geometry and the landscape includes [19].

2. Reflexive polytopes and their order

Let $M \cong \mathbb{Z}^d$ be a lattice, $N = \mathrm{Hom}(M,\mathbb{Z})$ its dual and $\langle \cdot,\cdot \rangle$ the pairing. For a polytope $P \subset M_{\mathbb{R}}$ with the origin in its interior the polar dual is $P^* = \{y \in N_{\mathbb{R}} : \langle x,y \rangle \geq -1 \text{ for all } x \in P\}$. A lattice polytope Δ is reflexive when Δ^* is a lattice polytope; then $\Delta^{**} = \Delta$, the facets of Δ are the sets $\{\langle x,v \rangle = -1\} \cap \Delta$ for the vertices v of Δ^*, and the origin is the only interior lattice point. For a reflexive Δ let $R(\Delta)$ be the set of reflexive subpolytopes of Δ.

LEMMA 7. *Let $\Gamma \subsetneq \Delta$ be lattice polytopes with the origin in their interiors. Then*

(i) $|\Gamma| < |\Delta|$;
(ii) $\Delta^* \subsetneq \Gamma^*$, *and if both are reflexive then* $|\Delta^*| < |\Gamma^*|$;
(iii) $R(\Delta)$ *is finite, and every element of* $R(\Delta)$ *other than* Δ *lies in a cover of* Δ.

Proof. (i) If every vertex of Δ lay in Γ, the convex hull Δ would lie in Γ; so some vertex of Δ, a lattice point, is missing from Γ. (ii) Polarity reverses inclusion and is injective on polytopes with the origin inside, and (i) applies to the lattice polytopes $\Delta^* \subsetneq \Gamma^*$. (iii) A reflexive subpolytope is the convex hull of a set of lattice points of Δ, so there are at most $2^{|\Delta|}$ of them. Given $\Gamma \in R(\Delta)$ with $\Gamma \neq \Delta$, a maximal element of $\{\Gamma' \in R(\Delta) : \Gamma \subseteq \Gamma' \neq \Delta\}$ exists by finiteness and is a cover of Δ. □

A polytope is r-minimal exactly when it has no cover. By Lemma 7(iii) a reflexive polytope that is not r-minimal has at least one cover, and we will need all of them.

3. The literal conjecture and reflexive descent

Proof of Theorem 2. Let Δ be reflexive. If Δ is r-minimal the descent of length zero will do. Otherwise Lemma 7(iii) gives a cover Δ_1, and we repeat with Δ_1. By Lemma 7(i) the number of lattice points drops at each step, so the process stops, and it stops exactly at an r-minimal polytope. Any reflexive descent ends at an r-minimal polytope by definition.

Let Δ_k be r-minimal and let V be its vertex set. Remove vertices from V one at a time, keeping the origin in the interior of the convex hull of what is left, until no vertex can be removed. The hull S of the remaining set is a minimal polytope, its vertices are vertices of Δ_k, and $S \subseteq \Delta_k$. If $S \neq \Delta_k$ the step $\Delta_k \supsetneq S$ leaves no room for a reflexive polytope between the two, because Δ_k has no reflexive subpolytope other than itself. Appending this step to a reflexive descent gives the chain of the theorem. □

Three readings of Conjecture 1 are now settled at once. If a "subpolytope step" may be any inclusion, the chain $\Delta \supsetneq S$ of the second paragraph applied to Δ itself has no intermediate hull at all, and $D = 1$; this is the minimal-polytope theorem of Kreuzer and Skarke [4] restated. If the seeds are the r-minimal polytopes, which are the seeds of the four-dimensional classification [6], a reflexive descent ends at a seed. If the seeds are minimal polytopes, at most

one further step is needed. In every reading the chain length is controlled by the transition depth, which is the quantity bounded in Theorem 3.[2]

The correcting note [18] stated that descent through reflexive polytopes only is not always possible. Theorem 2 shows that it always is. What fails is descent by deleting vertices, which is the operation the note had in mind, and it fails in two ways.

PROPOSITION 8. (i) *If Δ is a reflexive simplex, the convex hull of any proper subset of its vertices is a face of Δ and contains no interior point; so a reflexive simplex that is not r-minimal admits no descent by convex hulls of vertex subsets, although it has a reflexive descent.*

(ii) *Among the* 4,303 *reflexive three-polytopes that are not r-minimal, exactly* 74 *have no reflexive subpolytope of the form* $\operatorname{conv}\big((\Delta \cap M) \setminus \{v\}\big)$ *with v a vertex, and exactly* 354 *have none of the form* $\operatorname{conv}(V \setminus \{v\})$ *with V the vertex set.*

Proof. Part (i) is immediate: a proper subset of the vertices of a simplex spans a face of positive codimension. Part (ii) is a finite check on the list of Kreuzer and Skarke, recorded in the code archive [22]. □

Example 9. The smallest polytope in part (ii) has 15 lattice points and vertices $(1,0,0)$, $(0,1,0)$, $(-1,-1,0)$, $(0,0,1)$, $(-3,2,-2)$, $(-2,3,-2)$, $(-4,4,-3)$. Removing any one vertex and taking the hull of the remaining fourteen lattice points gives a polytope that is not reflexive, yet the polytope has a reflexive descent of length 5. In dimension four the largest polytope Δ_{680} is a simplex, so by part (i) it cannot be reduced by vertex deletion at all, while Theorem 5 gives it a descent of length 31.

The vertex-deletion chain of the note does reach a minimal polytope, and its length is less than the number of vertices [18]; in dimension four the vertex counts range from 5 to 33 [6], so such chains have at most 28 steps. Its intermediate hulls are not reflexive in general, which is why the conjecture was stated with reflexive intermediates.

[2] In dimensions two and three we checked which r-minimal polytopes are themselves minimal. The three r-minimal polygons are minimal; in dimension three, twelve of the sixteen r-minimal polytopes are minimal and four contain a smaller minimal polytope that is not reflexive. In dimension four Kreuzer and Skarke found seven r-minimal polytopes that do not come from weight systems, and each of them contains a single minimal subpolytope, a simplex that is not reflexive [6].

4. Counting lattice points

The lattice points of a polytope and of its dual move in opposite directions along a descent, and in dimension four both counts are known for every polytope of the list. This is enough to bound the length of every descent. We write $\mathcal{P}_4 \subset \mathbb{Z}^2$ for the set of pairs $(|\Delta|, |\Delta^*|)$ with $\Delta \in \mathrm{KS}_4$; both numbers are invariant under unimodular maps, and $\mathcal{P}_4$ is symmetric under $(p, q) \mapsto (q, p)$ because the list is closed under duality.

Definition 10. For $(p, q) \in \mathcal{P}_4$ let $L(p, q)$ be the largest k for which there are pairs $(p_0, q_0), \dots, (p_k, q_k)$ in $\mathcal{P}_4$ with $(p_0, q_0) = (p, q)$, $p_0 > p_1 > \cdots > p_k$ and $q_0 < q_1 < \cdots < q_k$. We call $L(p, q)$ the *pair bound* and write $L(\Delta) = L(|\Delta|, |\Delta^*|)$.

The pair bound satisfies $L(p, q) = 1 + \max L(p', q')$ over the pairs with $p' < p$ and $q' > q$, and $L(p, q) = 0$ when there is no such pair. Evaluating this recursion in increasing order of p computes L on all of $\mathcal{P}_4$ at once.

Lemma 11. *Let $\Delta_0 \supsetneq \Delta_1 \supsetneq \cdots \supsetneq \Delta_k$ be reflexive polytopes of dimension four, not necessarily a descent. Then*

$$k \;\le\; L(\Delta_0) \;\le\; \min\bigl(|\Delta_0| - 6,\; 680 - |\Delta_0^*|\bigr).$$

Proof. By Lemma 7 the counts $|\Delta_i|$ strictly decrease and the counts $|\Delta_i^*|$ strictly increase. Each Δ_i is unimodularly equivalent to a member of KS_4, so each pair $(|\Delta_i|, |\Delta_i^*|)$ lies in $\mathcal{P}_4$, and the pairs form a chain as in Definition 10. Hence $k \le L(\Delta_0)$. In any such chain of pairs the first coordinates are distinct integers between p_k and p_0 and the second are distinct integers between q_0 and q_k. Every reflexive polytope of dimension four has at least six lattice points and at most 680 [6], which gives the second inequality.[3] □

The set $\mathcal{P}_4$ was computed by reading the two point counts of all 473,800,776 polytopes of the list [12]. It has 71,870 elements, its extreme values 6 and 680 agree with those recorded by Kreuzer and Skarke [6], and the counts add up to the size of the list. Table 1 and Fig. 1 summarise both bounds over the list.

Proof of Theorem 3. Lemma 11 applies to every reflexive descent. Over $\mathcal{P}_4$ the largest value of $\min(p - 6, 680 - q)$ is 654, attained only at $(680, 26)$, and this pair belongs to one polytope, the simplex Δ_{680}; the next values are 634 at $(640, 26)$ and 633 at $(639, 27)$. The largest value of L is 348, attained

[3] The lower bound six is elementary: a full-dimensional lattice polytope in dimension four with an interior point has at least five vertices, and the origin is a further lattice point. The upper bound 680 is attained only by the simplex Δ_{680} of Section 7, as the list shows; Balletti, Kasprzyk and Nill proved that this simplex also has the largest volume among reflexive four-polytopes [14].

Table 1. Number of polytopes in KS_4 whose bound on the length of a reflexive descent exceeds a given value, for the simple bound $\min(|\Delta| - 6,680 - |\Delta^*|)$ and for the pair bound $L(\Delta)$. The note below the table gives the largest value of each bound and the point pair where it is attained.

bound exceeds	$\min(\lvert\Delta\rvert - 6,680 - \lvert\Delta^*\rvert)$	$L(\Delta)$
100	7,159,636	6,838,181
150	893,324	825,559
200	136,519	117,767
250	30,306	23,814
300	7,755	3,556
320	4,763	1,028
340	2,705	106
348	2,197	0
500	39	0
654	0	0

simple bound 654 at $(680, 26)$, the simplex Δ_{680}; pair bound 348 at $(521, 22)$, a single polytope with seven vertices.

only at $(521, 22)$, again by a single polytope, with vertices

$$(1,0,0,0),\ (0,1,0,0),\ (0,0,1,0),\ (1,1,1,2),$$
$$(-3,-3,1,-2),\ (-45,-38,-6,-30),\ (15,22,-6,30)$$

and Hodge numbers $(h^{1,1}, h^{2,1}) = (10, 376)$ for the hypersurface of which it is the Newton polytope. So every reflexive descent in dimension four has at most 348 steps, $D_4 \le 348$, and with one more step to a minimal polytope (Theorem 2) Conjecture 1 holds with $D = 349$. □

The pair bound of Δ_{680} is 347. A longest chain of pairs from $(521, 22)$ runs close to the upper edge of the occupied region and ends at the mirror pair $(22, 521)$ (Fig. 1). The chain is a property of the point counts only; nothing in the argument says that a descent follows it, and the descents we can compute are much shorter. For the polytope at $(521, 22)$, two covers found by the method of the next section lead to a polytope with 257 lattice points and 24 dual points, whose pair bound is 247, so that polytope has a descent of length at most 249.

Corollary 12. *The maximum shortest reflexive-descent depth satisfies* $22 \le D_4 \le 347$.

Proof. The unique polytope whose pair bound is 348 has a chain of two covers with point-count pairs $(521, 22)$, $(516, 23)$, $(257, 24)$. The final pair

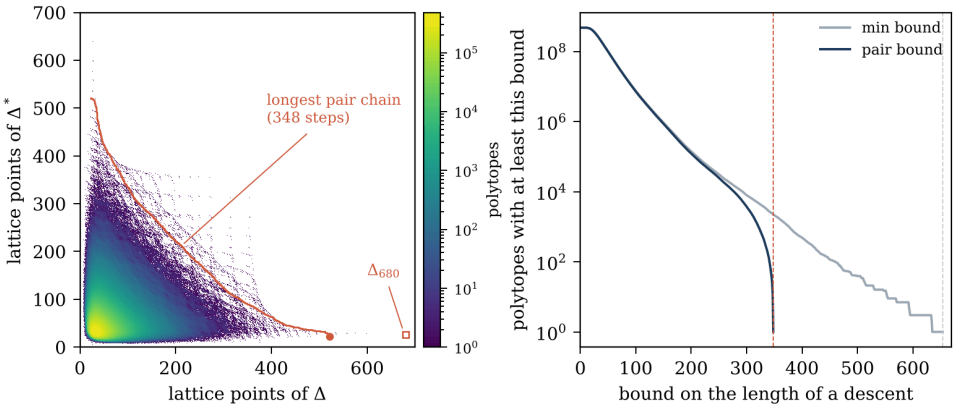


Figure 1. Left: the number of reflexive four-polytopes with a given number of lattice points (horizontal) and dual lattice points (vertical), on a logarithmic colour scale. A reflexive descent moves left and up through occupied cells. The red staircase is a longest chain of occupied cells, with 348 steps from $(521, 22)$ to its mirror image $(22, 521)$; the dot marks its start and the square marks the simplex Δ_{680} at $(680, 26)$. Right: for each value X, the number of polytopes whose bound on the length of a descent is at least X, for the simple bound and the pair bound.

has bound 247, so this polytope has shortest depth at most $2 + 247 = 249$. Every other polytope has pair bound at most 347. The lower bound 22 is unchanged. The certificate and an integer-arithmetic verifier are `code/data/top347_certificate.json` and `code/checks/check_top347.py`. This sharpens the bound for shortest descents; the bound 348 for arbitrary descents is unchanged. □

4.1. *Twelve interval certificates for shorter descents.*

LEMMA 13. *For reflexive polytopes $Q \subsetneq P$ in a fixed lattice,*

$$\delta(P) \leq \min\{|P| - |Q|,\ |Q^*| - |P^*|\} + \delta(Q).$$

In dimension four the last term can be replaced by the pair bound $L(|Q|, |Q^*|)$. The inclusion $Q \subsetneq P$ need not be a cover.

Proof. Choose a maximal chain in the finite interval $[Q, P]$ of reflexive subpolytopes. Any reflexive polytope strictly between consecutive members would also belong to that interval, so each step is a cover in the whole reflexive inclusion order. At each step at least one primal point is lost and at least one dual point is gained, by Lemma 7. The chain length is bounded by both

differences. Concatenate it with a shortest descent of Q. The pair-bound version uses Lemma 11 for the final descent. □

COROLLARY 14 (Finite-census refinement). *Relative to the supplied Kreuzer–Skarke point-pair census and the published classification, the maximum shortest depth satisfies* $22 \le D_4 \le 346$. *The arbitrary-descent bound remains* 348.

Proof. Compute $L(p,q) = \max(\{0\} \cup \{1 + L(p',q') : p' < p,\ q' > q\})$ over the occupied pairs, batching updates with equal p. Exactly twelve census entries have $L > 346$. The following strict reflexive intervals cover these entries; c is the minimum of the two point-count differences in Lemma 13.

$(\lvert P\rvert, \lvert P^*\rvert)$	$(\lvert Q\rvert, \lvert Q^*\rvert)$	c	$L(Q)$	$c + L(Q)$
$(445, 17)$	$(112, 22)$	5	105	110
$(680, 26)$	$(422, 27)$	1	339	340
$(640, 26)$	$(400, 27)$	1	338	339
$(600, 26)$	$(378, 27)$	1	326	327
$(560, 26)$	$(356, 27)$	1	325	326
$(576, 26)$	$(360, 27)$	1	326	327
$(520, 26)$	$(334, 27)$	1	315	316
$(520, 26)$	$(325, 27)$	1	308	309
$(516, 23)$	$(181, 25)$	2	173	175
$(521, 22)$	$(181, 25)$	3	173	176
$(536, 26)$	$(338, 27)$	1	316	317
$(482, 22)$	$(168, 25)$	3	161	164

The stored vertices and integral cut vectors specify each interval. The exact checker reconstructs every facet and every lattice point of P, Q, P^*, Q^*, verifies reflexivity and strict containment, and checks that the cut yields exactly the child's point set. It verifies the exceptional multiset against the census. The two representatives at $(520, 26)$ have different multisets of absolute four-by-four vertex determinants, proving that they are not unimodularly equivalent; the other ten representatives have distinct point pairs. Thus the twelve representatives exhaust the exceptional classes under the supplied census counts. All displayed bounds are below 346; every remaining pair has $L \le 346$. The earlier depth-22 certificate supplies the lower bound. □

The certificate records the histogram digest, vertices, primal points, cut vectors and coordinate-source digests. The verifier uses integer arithmetic and does not assert that these cuts are covers. Its output was rechecked for this revision. It uses the supplied histogram; it does not rederive the classification or rescan all 473,800,776 coordinate records. This distinction is separate from the completely analytic uniform bound and free-sum construction.

In other dimensions the joint distribution of point counts is not known, but the extremes are, and they already bound the depth. Let $s_1 = 2$ and $s_{k+1} = s_k^2 - s_k + 1$ be Sylvester's sequence, so that $s_2, s_3, s_4, s_5 = 3, 7, 43, 1807$.

PROPOSITION 15. *Let V_d be the largest normalised volume of a reflexive polytope of dimension d. Every chain $\Delta_0 \supsetneq \Delta_1 \supsetneq \cdots \supsetneq \Delta_k$ of reflexive polytopes of dimension d has*

$$k \;\leq\; V_d - 2.$$

In particular $D_d \leq V_d - 2$ for every $d \geq 2$, and $D_d \leq 2(s_d - 1)^2 - 2$ for every $d \geq 4$.

Proof. We include the lattice-point argument rather than leaving Blichfeldt's inequality [15] as an additional input. Triangulate a full-dimensional lattice polytope P with every point of $P \cap M$ as a vertex. Such a triangulation is obtained by an initial triangulation and successive stellar insertions of the remaining finite set of points. Let t be its number of full-dimensional simplices. Their facet-adjacency graph is connected: a generic path in the interior between two simplex interiors crosses only their facets. Order the simplices by traversing a spanning tree of this graph. The first simplex has $d + 1$ vertices, and each later simplex shares a facet with one already listed, adding at most one vertex. Thus $|P| \leq d + t$. Each full-dimensional lattice simplex has positive integral normalized volume, so $t \leq \mathrm{nvol}(P)$. Consequently $|P| \leq d + \mathrm{nvol}(P)$.

A full-dimensional lattice polytope with zero strictly inside has at least $d + 1$ vertices and the origin as a further lattice point, hence at least $d + 2$ points. Every proper lattice subpolytope omits a vertex of its parent, so each strict inclusion loses at least one lattice point. It follows that

$$k \leq |\Delta_0| - |\Delta_k| \leq (V_d + d) - (d + 2) = V_d - 2.$$

Finally Corollary 1.2 of the published volume theorem of Balletti, Kasprzyk and Nill [14] bounds the normalized volume of every reflexive d-polytope, $d \geq 4$, by $2(s_d - 1)^2$. This proves the dimension-only bound without using the Kreuzer–Skarke classification. □

COROLLARY 16 (An explicit terminating construction). *Given the vertices of a reflexive d-polytope P, there is a finite exact algorithm producing a saturated descent to an r-minimal seed. For $d = 4$ every output has at most 3526 steps, independently of any census. If a vertex-minimal seed rather than an r-minimal seed is required, append a vertex-deletion descent inside the last polytope; replace that auxiliary computation by a single final inclusion into its vertex-minimal endpoint. The same coarse bound applies to the resulting total number of strict inclusions. The appended endpoint need not be reflexive.*

Proof. List the lattice points of P in its finite coordinate bounding box. For each subset form its convex hull by exact rational arithmetic, test full dimension and strict containment of zero, and test that all primitive integral facet equations have constant term one. The retained hulls are exactly $R(P)$. At the current hull choose an inclusion-maximal proper retained subpolytope if one exists. This is a lower cover, since any intermediate would also occur in the list. Continue until there is no such hull. Point counts strictly decrease, so the procedure ends at an r-minimal polytope. No efficiency claim is made for this algorithm.

If needed, compute a vertex-minimal seed S inside the last polytope by repeatedly deleting a vertex while keeping zero interior to the full-dimensional hull. Retain only the final S and append the one inclusion into S, if proper, to the reflexive descent. Thus every member before the endpoint is reflexive; the auxiliary vertex-deletion hulls are not inserted into that chain. The same lattice-point proof bounds the resulting chain by $V_d - 2$. Since $s_4 = 43$, the explicit four-dimensional bound is $2 \cdot 42^2 - 2 = 3526$. □

The bound is doubly exponential in d, and in low dimensions it is weak: $V_2 = 9$ and $V_3 = 72$ give $D_2 \leq 7$ and $D_3 \leq 70$ against the exact values 5 and 11 of Theorem 4, and in dimension four it gives $D_4 \leq 3526$ against 348. Its point is that it needs no classification. The dual side gives nothing better: the counts $|\Delta_i^*|$ rise from at least $d + 2$ to at most $V_d + d$, which is the same bound.

5. **Covers as cuts**

To compute transition depths we need all covers of a reflexive polytope. In dimension two they are easy to list (Section 6), but in dimension four a polytope can have hundreds of lattice points and enumerating subsets is out of the question. The covers turn out to be cuts by single dual lattice points.

For a reflexive Δ and $q \in N$ put

$$H_q = \operatorname{conv}\{x \in \Delta \cap M : \ \langle x, q\rangle \geq -1\},$$

the part of Δ that survives the cut by the hyperplane $\langle x, q\rangle = -1$.

Lemma 17 (Cut lemma). *Let $\Gamma \subseteq \Delta$ be reflexive polytopes and q a lattice point of Γ^*. Then $\Gamma \subseteq H_q$.*

Proof. Every lattice point x of Γ lies in Δ and satisfies $\langle x, q\rangle \geq -1$ because $q \in \Gamma^*$, so it is among the points whose convex hull is H_q; and Γ is the convex hull of its lattice points. □

For a vertex v of Δ let $F_v = \{y \in \Delta^* : \langle v, y\rangle = -1\}$ be the dual facet and $w_v = \operatorname{nvol}\operatorname{conv}(\{0\} \cup F_v)$ its normalised volume seen from the origin, so that $\operatorname{nvol}\Delta^* = \sum_v w_v$. Let V_d be the largest normalised volume of a reflexive

polytope of dimension d. The classification gives $V_2 = 9$ and $V_3 = 72$, and Balletti, Kasprzyk and Nill proved $V_4 = 2 \cdot 42^2 = 3528$, attained by Δ_{680} [14].

PROPOSITION 18. *Let Γ be a cover of the reflexive polytope $\Delta \subset M_{\mathbb{R}}$ of dimension d, and let q be a vertex of Γ^* that does not lie in Δ^*. Then*

(a) *$q \in N$, and $\langle v, q\rangle \le -2$ for some vertex v of Δ;*
(b) *$\langle x, q\rangle = -1$ for d linearly independent lattice points x of Δ;*
(c) *$\sum_v w_v \max(1, -\langle v, q\rangle) \le V_d$, the sum running over the vertices of Δ;*
(d) *$\Gamma \subseteq H_q \subsetneq \Delta$, and either $H_q = \Gamma$, or H_q is not reflexive and Γ is a maximal reflexive polytope inside it.*

Proof. Since Γ is reflexive, Γ^* is a lattice polytope and its vertex q lies in N. As $q \notin \Delta^* = \{y : \langle v, y\rangle \ge -1 \text{ for all vertices } v\}$, some vertex has $\langle v, q\rangle < -1$, hence ≤ -2. This is (a). The vertex q of Γ^* corresponds to a facet of Γ lying on the hyperplane $\langle x, q\rangle = -1$. That facet is a lattice polytope of dimension $d-1$, so it has d affinely independent lattice points; they are linearly independent because the hyperplane misses the origin, and they are lattice points of Δ. This is (b).

For (c), Γ^* is a convex set containing Δ^* and q, so it contains $C = \operatorname{conv}(\Delta^* \cup \{q\})$. The polytope C is the union of Δ^* and the pyramids with apex q over the facets F_v visible from q, that is, those with $\langle v, q\rangle < -1$, and these pieces meet only along their boundaries. The lattice distance from q to the hyperplane of F_v is $-\langle v, q\rangle - 1$, while the distance from the origin is 1, so the pyramid over F_v has normalised volume $w_v(-\langle v, q\rangle - 1)$. Adding $\operatorname{nvol}\Delta^* = \sum_v w_v$ gives $\operatorname{nvol} C = \sum_v w_v \max(1, -\langle v, q\rangle)$. Since Γ^* is reflexive, $\operatorname{nvol} C \le \operatorname{nvol}\Gamma^* \le V_d$.

For (d), the cut lemma gives $\Gamma \subseteq H_q$. The vertex v of (a) is a lattice point of Δ that is not in H_q, so $H_q \subsetneq \Delta$. If H_q is reflexive, it lies strictly between Γ and Δ or equals Γ, and the first is excluded because Γ is a cover. Otherwise any reflexive Γ' with $\Gamma \subseteq \Gamma' \subseteq H_q$ satisfies $\Gamma \subseteq \Gamma' \subsetneq \Delta$, so $\Gamma' = \Gamma$. □

Conditions (a) and (c) confine q to a bounded set: by (c) each $-\langle v, q\rangle$ is at most V_d/w_v, and the vertices of Δ span $M_{\mathbb{R}}$ positively. So only finitely many cuts have to be tried.

COROLLARY 19. *Let $\mathcal{C}(\Delta)$ be the set of reflexive polytopes obtained as follows: for every q satisfying (a), (b) and (c), take H_q if it is reflexive, and otherwise its maximal reflexive subpolytopes. The covers of Δ are exactly the maximal elements of $\mathcal{C}(\Delta)$ under inclusion.*

Proof. Every element of $\mathcal{C}(\Delta)$ is a reflexive polytope strictly inside Δ, and by Proposition 18 every cover belongs to $\mathcal{C}(\Delta)$. A cover is maximal among all reflexive proper subpolytopes, hence maximal in $\mathcal{C}(\Delta)$. Conversely, a maximal

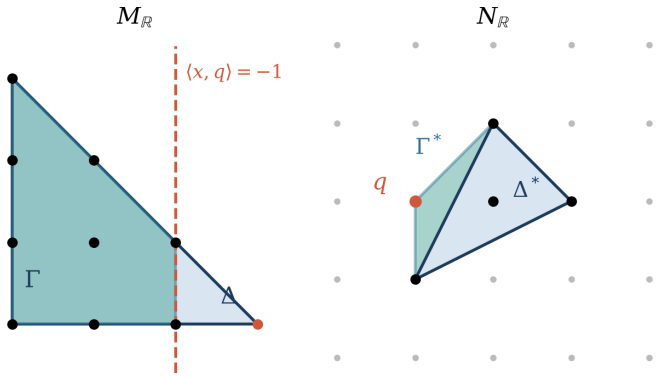


Figure 2. A cover as a cut. Left: a reflexive triangle Δ with ten lattice points and the cover Γ (shaded) obtained by removing the lattice points with $\langle x, q\rangle < -1$. Right: the duals; Γ^* is Δ^* with the new vertex q, and the added piece is the pyramid of Proposition 18(c) over the facet of Δ^* visible from q.

element X of $\mathcal{C}(\Delta)$ lies in some cover C by Lemma 7(iii), and $C \in \mathcal{C}(\Delta)$, so $X = C$. □

Fig. 2 shows a cut in dimension two. In dimension two the second case of Proposition 18(d) never occurs: H_q contains Γ, hence the origin in its interior, and lies in Δ, so the origin is its only interior lattice point, and a lattice polygon with exactly one interior lattice point is reflexive [16]. From dimension three on a lattice polytope with one interior lattice point need not be reflexive, and the argument fails. Yet the second case does not occur in dimension three either: over all 4,319 reflexive polyhedra, every cut H_q formed by the procedure of Corollary 19 is reflexive, and for each of the 32,129 pairs of a cover Γ and a vertex q of Γ^* outside Δ^* the cut H_q is Γ itself. In dimension four the same holds at each of the 5,250 polytopes of the certificate of Proposition 23 that have covers: every cut H_q formed by the procedure is reflexive, and for each of the 50,280 pairs of a cover Γ and a vertex q of Γ^* outside Δ^* the cut H_q is Γ itself. Whether the second case occurs anywhere in dimension four is open. The programs allow for the second case and then find the maximal reflexive subpolytopes by removing vertices one at a time.

Listing all covers is still expensive for the largest polytopes. To walk down from a polytope it is enough to certify one cover at a time, and for that the following lemma needs only the dual lattice points of the candidate.

Lemma 20 (Climbing lemma). *Let $\Gamma \subsetneq \Delta$ be reflexive. If $H_{q'} = \Gamma$ for every lattice point q' of Γ^* outside Δ^*, then Γ is a cover of Δ.*

Proof. Suppose $\Gamma \subsetneq \Gamma' \subsetneq \Delta$ with Γ' reflexive. Then $\Delta^* \subsetneq \Gamma'^* \subsetneq \Gamma^*$, so the lattice polytope Γ'^* has a vertex $q' \notin \Delta^*$, and q' is a lattice point of Γ'^* and of Γ^*. The cut lemma, applied to $\Gamma' \subseteq \Delta$ and q', gives $\Gamma' \subseteq H_{q'} = \Gamma$, a contradiction. □

The lemma turns into a procedure. Start from a cut q with H_q reflexive. For each lattice point q' of Γ^* outside Δ^*, the polytope $H_{q'}$ contains Γ by the cut lemma; if it is reflexive and larger, replace Γ by it and start again. Since the polytopes grow inside Δ, this stops. When it stops with $H_{q'} = \Gamma$ for every such q', the lemma certifies Γ as a cover; every step of the descent of Section 7 that is attributed to the climbing lemma was certified in this way. Choosing the first cut so that Γ is small gives descents that lose many points per step, which is how the descent of Section 7 was found.

6. **Dimensions two and three**

In dimension two covers are simple.

PROPOSITION 21. *If Γ is a cover of a reflexive polygon Δ, then $|\Gamma| = |\Delta|-1$. Every r-minimal polygon $E \subseteq \Delta$ is the last member of some reflexive descent of Δ, and the transition depth of a reflexive polygon is $\delta(\Delta) = |\Delta| - m(\Delta)$, where $m(\Delta)$ is the largest number of lattice points of an r-minimal polygon inside Δ.*

Proof. A lattice polygon with exactly one interior lattice point is reflexive [16]. Let $\Gamma \subsetneq \Delta$ be reflexive. Some vertex v of Δ is not in Γ, and $\Delta' = \operatorname{conv}\big((\Delta \cap M) \setminus \{v\}\big)$ contains Γ, hence the origin in its interior, and has no other interior lattice point; so Δ' is reflexive, contains Γ, and has exactly $|\Delta| - 1$ lattice points, since a vertex is not in the hull of the other lattice points. If Γ is a cover, then $\Gamma = \Delta'$.

The same argument, applied to any reflexive P with $E \subsetneq P \subseteq \Delta$ and a vertex of P outside E, gives a cover of P that still contains E, so repeating it gives a descent of Δ that ends at E, of length $|\Delta| - |E|$. Every descent loses one point per step and ends at an r-minimal polygon, which gives the formula. □

Proof of Theorem 4 in dimension two. There are sixteen reflexive polygons, listed in Appendix A with their covers; three are r-minimal, the square $\operatorname{conv}(\pm e_1, \pm e_2)$ and the triangle with vertices $(1,0)$, $(0,1)$, $(-2,-1)$ with five lattice points each, and the triangle with vertices $(1,0)$, $(0,1)$, $(-1,-1)$ with four. Every polygon with at least six lattice points contains one of the two r-minimal polygons with five points (the table names one), so by Proposition 21 its transition depth is $|\Delta| - 5$. The largest polygon is the triangle $\operatorname{conv}\{(-1,-1),(2,-1),(-1,2)\}$ with ten lattice points; it contains the square, so its depth is 5, and every other polygon has at most nine points and depth

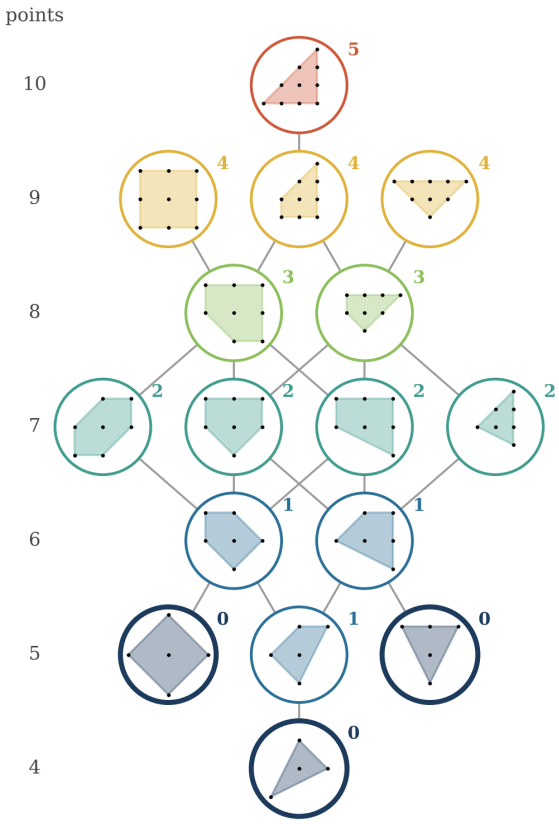


Figure 3. The sixteen reflexive polygons ordered by inclusion, by number of lattice points; edges are covers and the number beside each polygon is its transition depth. The three r-minimal polygons are drawn with thick circles.

at most 4. Hence $D_2 = 5$, attained only by this triangle. It also contains the triangle with four points, so its longest descent has 6 steps. □

Fig. 3 shows the whole order. The argument uses only that in the plane one interior point forces reflexivity; that fails from dimension three on, and there covers can drop many points at once.

In dimension three we computed the covers of all 4,319 reflexive polytopes in two independent ways. The first enumerates, for each polytope, all lattice polytopes obtained by repeatedly removing a vertex and taking the convex hull of the remaining lattice points, as long as the origin stays inside; up to unimodular equivalence this gives 665,599 polytopes, among them every reflexive subpolytope, and the covers are read off as maximal elements. The second uses Corollary 19 with $V_3 = 72$. The two agree on all 4,303 polytopes that are not r-minimal, and give 21,841 covers in all.

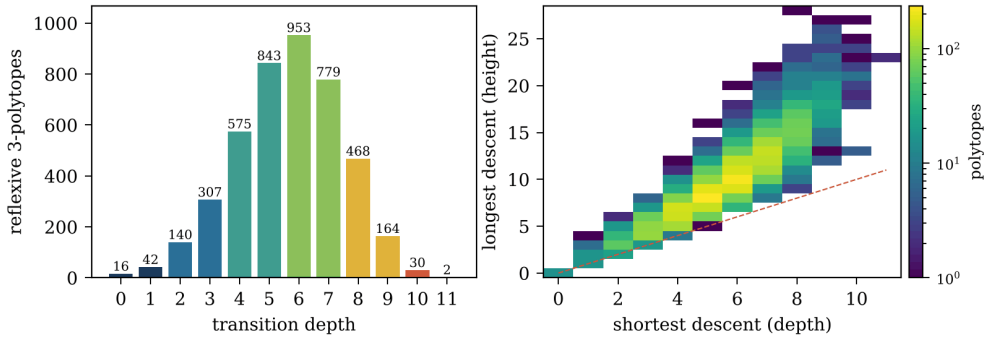


Figure 4. Reflexive three-polytopes. Left: the number of polytopes with each transition depth. Right: transition depth against the length of the longest descent, with the diagonal dashed.

Proof of Theorem 4 in dimension three. With the cover relation known, transition depths follow by induction on the number of lattice points: $\delta(\Delta) = 0$ if Δ has no cover and $\delta(\Delta) = 1 + \min \delta(\Gamma)$ over its covers otherwise. The maximum is 11, attained by exactly two polytopes. The first has 35 lattice points and vertices

$$(1,0,0),\ (1,2,0),\ (-3,-2,0),$$
$$(1,2,4),\ (-3,-2,-4),$$

and the second has 31 lattice points and vertices

$$(1,0,0),\ (0,1,0),\ (1,0,2),$$
$$(1,-4,-2),\ (-2,3,2),\ (-2,-1,-2).$$

Neither is contained in a larger reflexive polytope, and both have longest descents of 23 steps. □

The distribution of depths in dimension three is shown in Fig. 4. The longest descent of any three-polytope has 28 steps, against a simple counting bound of 33, and sixteen polytopes are r-minimal. Shortest and longest descents differ a great deal. The mean transition depth is 5.67, while the mean length of a longest descent is 10.23, so a well-chosen descent is on average a little more than half as long as a badly chosen one.

7. **Dimension four**

The largest reflexive four-polytope is the simplex

$$\Delta_{680} = \operatorname{conv}\{(-1,-1,-1,-1),\ (83,-1,-1,-1),$$
$$(-1,6,-1,-1),\ (-1,-1,2,-1),\ (-1,-1,-1,1)\}.$$

Table 2. The sixteen reflexive polygons. Here $|\Delta|$ is the number of lattice points, c the number of covers, δ the transition depth and h the length of a longest descent; the last column gives an r-minimal subpolygon with five lattice points. The three r-minimal polygons are the last two rows with five points and the row with four.

vertices of Δ	$\lvert\Delta\rvert$	c	δ	h	five-point r-minimal polygon inside Δ
$(-2,-3)$, $(1,0)$, $(1,3)$	10	1	5	6	$(-1,-2)$, $(0,1)$, $(1,0)$
$(-2,1)$, $(0,1)$, $(1,-2)$, $(1,0)$	9	2	4	5	$(-2,1)$, $(0,1)$, $(1,-1)$
$(-1,-2)$, $(-1,0)$, $(1,0)$, $(1,2)$	9	1	4	5	$(-1,-2)$, $(0,1)$, $(1,0)$
$(-3,-2)$, $(1,0)$, $(1,2)$	9	1	4	5	$(-2,-1)$, $(0,1)$, $(1,0)$
$(-1,-2)$, $(-1,1)$, $(0,1)$, $(1,0)$	8	3	3	4	$(-1,-2)$, $(0,1)$, $(1,0)$
$(-1,-1)$, $(-1,1)$, $(0,1)$, $(1,-1)$, $(1,0)$	8	3	3	4	$(-1,-1)$, $(0,1)$, $(1,-1)$
$(-3,-2)$, $(0,1)$, $(1,0)$	7	1	2	3	$(-2,-1)$, $(0,1)$, $(1,0)$
$(-1,-1)$, $(-1,1)$, $(0,-1)$, $(0,1)$, $(1,0)$	7	2	2	3	$(-1,-1)$, $(-1,1)$, $(1,0)$
$(-2,1)$, $(0,-1)$, $(0,1)$, $(1,0)$	7	2	2	3	$(-2,1)$, $(0,-1)$, $(1,0)$
$(-1,0)$, $(-1,1)$, $(0,-1)$, $(0,1)$, $(1,-1)$, $(1,0)$	7	1	2	3	$(-1,1)$, $(0,-1)$, $(0,1)$, $(1,-1)$
$(-1,-1)$, $(-1,1)$, $(0,1)$, $(1,0)$	6	2	1	2	$(-1,-1)$, $(-1,1)$, $(1,0)$
$(-1,0)$, $(0,-1)$, $(0,1)$, $(1,0)$, $(1,1)$	6	2	1	2	$(-1,0)$, $(0,-1)$, $(0,1)$, $(1,0)$
$(-1,0)$, $(0,1)$, $(1,-1)$, $(1,0)$	5	1	1	1	—
$(-2,-1)$, $(0,1)$, $(1,0)$	5	0	0	0	—
$(-1,0)$, $(0,-1)$, $(0,1)$, $(1,0)$	5	0	0	0	—
$(-1,-1)$, $(0,1)$, $(1,0)$	4	0	0	0	—

Table 3. The reflexive descent $\Gamma_0 = \Delta_{680} \supsetneq \Gamma_1 \supsetneq \cdots \supsetneq \Gamma_{38}$: lattice points of Γ_i and of Γ_i^*, and vertices of Γ_i.

i	0	1	2	3	4	5	6	7	8	9	10	11	12
$\lvert\Gamma_i\rvert$	680	422	236	207	137	122	103	78	57	44	37	36	35
$\lvert\Gamma_i^*\rvert$	26	27	28	29	30	32	33	34	35	36	37	38	39
vert.	5	7	7	9	11	9	7	9	7	7	5	6	6

i	13	14	15	16	17	18	19	20	21	22	23	24	25
$\lvert\Gamma_i\rvert$	34	33	32	31	30	29	28	27	26	25	24	23	22
$\lvert\Gamma_i^*\rvert$	40	42	44	46	48	51	54	58	62	66	70	80	86
vert.	6	7	7	7	7	7	7	7	7	7	7	8	8

i	26	27	28	29	30	31	32	33	34	35	36	37	38
$\lvert\Gamma_i\rvert$	21	20	19	18	17	16	15	14	13	12	11	10	9
$\lvert\Gamma_i^*\rvert$	88	92	104	116	128	140	152	164	179	188	196	200	201
vert.	7	7	8	8	8	8	8	7	7	7	7	6	5

Its lattice points are the exponents of the monomials of degree 84 in the weighted projective space $\mathbb{P}(1,1,12,28,42)$, shifted by the interior point $(1,1,1,1)$, so

it has 680 lattice points. The dual simplex has vertices e_1, e_2, e_3, e_4 and $(-1, -12, -28, -42)$, 26 lattice points and normalised volume $1+1+12+28+42 = 84$, and Δ_{680} has normalised volume $3528 = 2 \cdot 42^2$, the largest possible [14]. The weights divided by their sum are $\frac{1}{2}, \frac{1}{3}, \frac{1}{7}, \frac{1}{84}, \frac{1}{84}$, which add up to 1 because $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{42} = 1$; here $2, 3, 7, 43$ are the first terms of Sylvester's sequence and $42 = 43 - 1$. The hypersurfaces have Hodge numbers $(h^{1,1}, h^{2,1}) = (11, 491)$, and their mirror has the largest $h^{1,1}$ in the list.

Because Δ_{680} is a simplex, no proper subset of its vertices spans a polytope with the origin inside, so vertex deletion cannot even start (Proposition 8). Reflexive descent works (Fig. 5). Ten steps of the climbing procedure of Section 5 take the number of lattice points through

$$680,\ 422,\ 236,\ 207,\ 137,\ 122,\ 103,\ 78,\ 57,\ 44,\ 37,$$

each step certified as a cover by Lemma 20. From 37 points on, all covers were computed by Corollary 19; each step removes one point, and after 38 steps the descent reaches a simplex with nine lattice points and vertices

$$(-1, -1, -1, 1),\ (-1, -1, 2, -1),\ (-1, 0, 1, -1),$$
$$(-1, 6, -1, -1),\ (1, 0, 0, 0),$$

which has no reflexive subpolytope and is itself minimal. The dual counts rise from 26 to 201 along the way. Appendix B lists the whole descent.

7.1. *A certificate for lower bounds.* An upper bound for a transition depth needs one descent, a lower bound a statement about all of them. The following lemma reduces a lower bound to a finite object that can be checked one polytope at a time.

Lemma 22 (Certificate lemma). *Let $\mathcal{N}$ be a set of reflexive polytopes and $b\colon \mathcal{N} \to \mathbb{Z}_{\geq 0}$ a function such that for every $\Gamma \in \mathcal{N}$ with $b(\Gamma) > 0$ the polytope Γ has a cover, and every cover of Γ is unimodularly equivalent to some $\Gamma' \in \mathcal{N}$ with $b(\Gamma') \geq b(\Gamma) - 1$. Then $\delta(\Gamma) \geq b(\Gamma)$ for every $\Gamma \in \mathcal{N}$.*

Proof. A unimodular map fixes the lattice and the origin and preserves inclusion, so it carries covers to covers and r-minimal polytopes to r-minimal ones; hence δ is constant on classes of unimodularly equivalent polytopes. We use induction on $|\Gamma|$. If $b(\Gamma) = 0$ there is nothing to prove. Otherwise Γ has a cover, so it contains a reflexive polytope other than itself and is not r-minimal, and every descent $\Gamma = \Gamma_0 \supsetneq \Gamma_1 \supsetneq \cdots \supsetneq \Gamma_k$ has $k \geq 1$. Its first step Γ_1 is a cover of Γ, so it is equivalent to some $\Gamma' \in \mathcal{N}$ with $b(\Gamma') \geq b(\Gamma) - 1$, and $|\Gamma'| = |\Gamma_1| < |\Gamma|$ by Lemma 7(i). The chain $\Gamma_1 \supsetneq \cdots \supsetneq \Gamma_k$ is a descent of Γ_1, so the induction hypothesis gives $k - 1 \geq \delta(\Gamma_1) = \delta(\Gamma') \geq b(\Gamma') \geq b(\Gamma) - 1$, and $k \geq b(\Gamma)$. □

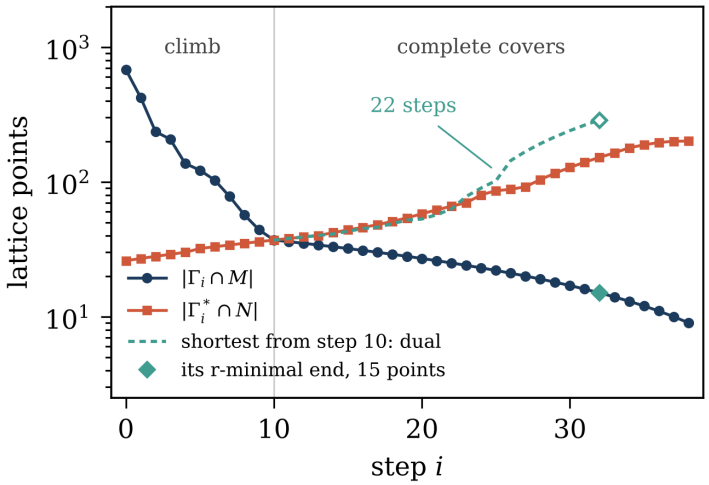


Figure 5. The descent from Δ_{680}: lattice points of Γ_i and of its dual at each step. The first ten covers were certified by the climbing lemma, the remaining ones by listing all covers. The polytope Γ_{10} is the polytope P_1 of Section 7.2; a shortest descent of it has 22 steps, removes one lattice point at each step like the descent drawn, and ends at an r-minimal polytope with 15 lattice points (filled diamond) whose dual has 288 (open diamond). The dashed line shows its dual counts.

The lemma asks for a complete list of covers only at the polytopes with positive bound, and Corollary 19 provides such lists. The polytopes need not be the same as those of any particular descent, and the bounds need not be exact depths. A certificate is produced by a bounded search: to show $\delta(\Gamma) > \ell$ it lists the covers of Γ and shows $\delta(\Gamma') > \ell - 1$ for each of them, identifying unimodularly equivalent polytopes by their normal forms and keeping every bound it has proved, so that a polytope reached along many paths is examined once.[4] The certificate is then checked by a program that does not repeat the search: it verifies the hypothesis of the lemma polytope by polytope and recomputes the lists of covers.

[4]The search is a depth-first search with a memory of exact depths and of lower bounds. For a bound ℓ it tries the covers of Γ in the order of their known depths, smallest first, and stops as soon as one of them has depth at most $\ell - 1$; if none has, the bound $\delta(\Gamma) \geq \ell + 1$ is recorded. Repeating with $\ell = 0, 1, 2, \ldots$ gives the exact depth, and the recorded bounds of all polytopes that were reached form a function b as in Lemma 22. Exact depths found on the way are recorded as well; they give the descents that prove the matching upper bounds.

7.2. *Two polytopes of depth twenty-two.* Let $e_1, \dots, e_4$ be the standard basis of $M = \mathbb{Z}^4$, put $u = (-4, -2, -1, 0)$, and let

$$P_1 = \operatorname{conv}\{e_1,\ e_2,\ e_3,\ u + 8e_4,\ u - 8e_4\},$$
$$P_2 = \operatorname{conv}\{e_1,\ e_2,\ e_3,\ u,\ u + 8e_4,\ e_3 - 8e_4\}.$$

Both polytopes contain $Q \times \{0\}$, where the tetrahedron $Q = \operatorname{conv}\{e_1, e_2, e_3, (-4, -2, -1)\}$ lies in $\mathbb{R}^3$ with the lattice $\mathbb{Z}^3$, and both are bipyramids over it with apexes at height ± 8: P_1 is the convex hull of $Q \times \{0\}$ and the two points $((-4, -2, -1), \pm 8)$, and P_2 that of $Q \times \{0\}$ and the points $((-4, -2, -1), 8)$ and $(e_3, -8)$. The tetrahedron Q is reflexive, it is the simplex of the weights $(4, 2, 1, 1)$ because $4e_1 + 2e_2 + e_3 + (-4, -2, -1) = 0$, and it has the seven lattice points

$$e_1,\ e_2,\ e_3,\ (-4, -2, -1),\ (-2, -1, 0),\ (-1, 0, 0),\ 0.$$

It contains no reflexive polytope other than itself, so it is r-minimal and its transition depth in dimension three is 0.

The polytope P_1 is a simplex, and $8e_1 + 4e_2 + 2e_3 + (u + 8e_4) + (u - 8e_4) = 0$, so it is the simplex of the weights $(8, 4, 2, 1, 1)$. Its lattice points lie on vertical lines over the lattice points of Q. A point (x, t) with $x \in Q$ lies in P_1 exactly when $|t| \le 8\lambda(x)$, where $\lambda(x)$ is the barycentric coordinate of x in Q at the vertex $(-4, -2, -1)$; indeed, writing $(x, t) = \mu_1 e_1 + \mu_2 e_2 + \mu_3 e_3 + \alpha(u + 8e_4) + \beta(u - 8e_4)$ with nonnegative coefficients of sum 1 gives $\alpha + \beta = \lambda(x)$ and $t = 8(\alpha - \beta)$. The barycentric coordinates of the four lattice points $(-4, -2, -1)$, $(-2, -1, 0) = \frac{1}{2}(-4, -2, -1) + \frac{1}{2}e_3$, $(-1, 0, 0) = \frac{1}{4}(-4, -2, -1) + \frac{1}{2}e_2 + \frac{1}{4}e_3$ and $0 = \frac{1}{8}(-4, -2, -1) + \frac{1}{2}e_1 + \frac{1}{4}e_2 + \frac{1}{8}e_3$ at that vertex are $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$, so the lines over them carry 17, 9, 5 and 3 lattice points, and the three points e_1, e_2, e_3 carry one each (Fig. 6(a)). Hence

$$|P_1| = 17 + 9 + 5 + 3 + 3 = 37.$$

The dual simplex P_1^* has the vertices $(-1, -1, -1, \pm 1)$, $(-1, -1, 7, 0)$, $(-1, 3, -1, 0)$ and $(1, -1, -1, 0)$, read off from the five facets of P_1, and it also has 37 lattice points. The polytope P_2 has six vertices, and $|P_2| = |P_2^*| = 37$. For both polytopes the Calabi–Yau hypersurfaces have Hodge numbers $(h^{1,1}, h^{2,1}) = (34, 34)$.

The two polytopes were found by a search over skew bipyramids $\operatorname{conv}(Q' \times \{0\} \cup \{(a, k), (b, -k)\})$ over three-dimensional reflexive polytopes Q', with a, b lattice points of Q' and $k \ge 1$, ranked by the length of a descent found greedily with the cuts of Section 5; the exact depths of the highest ranked were then computed with complete lists of covers.[5]

[5]The search was organised on the Anaxia engine of the BLLM programme [20] and [21], release 3.0.5, its whole record was rerun on release 3.1.1, and it is kept in release 3.2.1. Anaxia

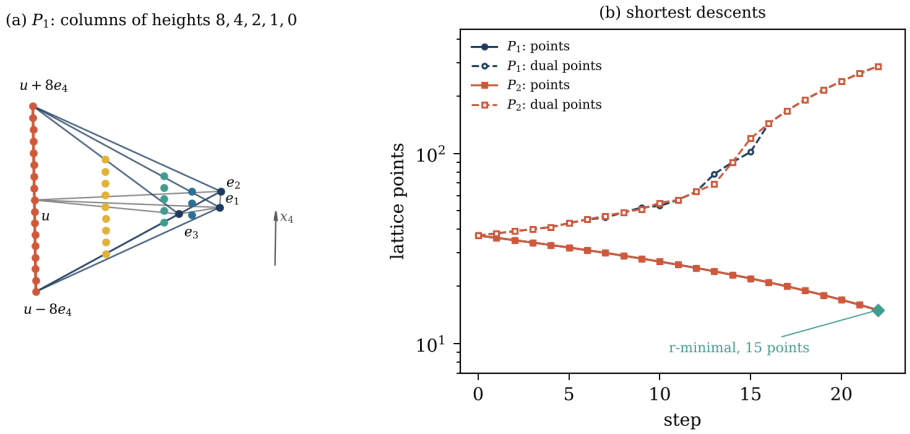


Figure 6. (a) The polytope P_1 as a bipyramid over the tetrahedron $Q \times \{0\}$ (grey edges), with the fourth coordinate vertical. Its 37 lattice points lie on vertical lines over the seven lattice points of Q, with 17, 9, 5 and 3 points over $(-4,-2,-1)$, $(-2,-1,0)$, $(-1,0,0)$ and 0 and one point at each of e_1, e_2, e_3; the half-lengths $8, 4, 2, 1, 0$ of these lines are 8 times the barycentric coordinate at the vertex $(-4,-2,-1)$ of Q. (b) Lattice points (solid) and dual lattice points (dashed) along shortest descents of P_1 and P_2. Each step removes one lattice point, and both descents end at the same r-minimal polytope, with 15 lattice points and 288 dual lattice points (diamond).

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runs a failure-driven loop: each strategy for a target is a list of checks, a computation or a proof with a named source, a failed check is recorded together with the change of strategy it forces, all checks are rerun from the archived data, and the label of a target (open, candidate, proved) is computed from the record, never assigned by hand. For the upper bound of Theorem 3 its record shows the first two strategies failing: descent by deleting one vertex at a time fails on 7 of 400 sampled three-dimensional polytopes, which have no reflexive vertex deletion, and cuts at dual lattice points of gauge two miss covers on 73 of the 400; the change forced by the second failure was the complete set of cuts of Corollary 19, which agrees with the vertex-deletion order on all of them. For the sharp form of Conjecture 1 the first strategy, a sweep of the whole list, failed on cost, since 162,328,050 polytopes have $\min(|\Delta| - 6, 680 - |\Delta^*|) > 40$; the second and third gave the descent of Δ_{680} with exact depths of small polytopes along it; the fourth was the search over skew bipyramids that produced P_1 and P_2, with the certificate of Proposition 23 and the identification of Proposition 24 as its checks. Its check that the sharp constant is determined fails, and that target stays open. The engine organises and reruns the computations; every statement of this section is proved in the text, and the certificate is checked by the programs of Appendix C, which do not use the engine.

PROPOSITION 23. $\delta(P_1) = \delta(P_2) = 22$.

Proof. For the upper bound, each of P_1 and P_2 has a descent of 22 steps, each step a cover by Corollary 19. Both remove one lattice point at each step, so the numbers of lattice points run through $37, 36, \ldots, 15$, while the dual counts rise from 37 to 288 (Fig. 6(b)). Both end at the same polytope, with vertices

$$(-4,-2,-1,7),\ (-2,-1,0,-1),\ (-2,-1,0,4),\ e_1,\ e_2,\ e_3,$$

15 lattice points and no cover.

For the lower bound we apply Lemma 22 to a set $\mathcal{N}$ of 5,592 polytopes containing P_1 and P_2, with $b(P_1) = b(P_2) = 22$. Of these, 5,250 have positive bound; for each of them the set $\mathcal{N}$ contains a representative of every cover, given by the complete list of Corollary 19, 23,646 covers in all, and each cover has bound at least one less. The lemma gives $\delta(P_1), \delta(P_2) \geq 22$. □

The polytopes of the certificate have between 14 and 37 lattice points; the most frequent sizes are 17 to 20 points, with between 472 and 494 polytopes each, and only 61 polytopes have more than 31 points.

PROPOSITION 24. *The matrix*

$$G = \begin{pmatrix} -1 & -1 & -1 & -1 \\ -1 & -1 & 6 & 0 \\ -1 & 2 & -1 & 0 \\ 1 & -1 & -1 & 0 \end{pmatrix}$$

is unimodular, and $x \mapsto Gx$ carries P_1 onto the polytope Γ_{10} of the descent of Δ_{680} in Table 3. Hence $\delta(\Gamma_{10}) = 22$ and $\delta(\Delta_{680}) \leq 32$.

Proof. The matrix

$$H = \begin{pmatrix} 0 & 3 & 7 & 11 \\ 0 & 2 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ -1 & -6 & -14 & -21 \end{pmatrix}$$

is integral and $GH = 1$, so G is unimodular. The map sends e_1, e_2, e_3, $u + 8e_4$ and $u - 8e_4$ to

$$(-1,-1,-1,1),\ (-1,-1,2,-1),\ (-1,6,-1,-1),$$
$$(-1,0,1,-1),\ (15,0,1,-1),$$

which are the five vertices of the simplex Γ_{10}. Since Γ_{10} is reached from Δ_{680} by ten covers, $\delta(\Delta_{680}) \leq 10 + \delta(\Gamma_{10}) = 10 + 22$. □

Three of the vertices of Γ_{10} are vertices of Δ_{680}, and the other two lie on its facet $x_4 = -1$. One step higher the same search gives a descent of 22 steps from

Γ_9, which has 44 lattice points: it removes one or two points at a time and ends at an r-minimal polytope with 19 lattice points. Hence $\delta(\Delta_{680}) \le 9 + 22 = 31$.

Proof of Theorem 5. Proposition 23 gives $\delta(P_1) = \delta(P_2) = 22$, so $D_4 \ge 22$, and Proposition 24 identifies P_1 with Γ_{10}. The nine covers from Δ_{680} to Γ_9 in Table 3, each certified by the climbing lemma, followed by the descent of 22 steps from Γ_9, form a descent of Δ_{680} of length 31. Finally $D_4 \le 348$ by Theorem 3. □

Smaller polytopes along the descent of Δ_{680} have smaller depth: the same search gives $\delta(\Gamma_{27}) = 9$ for the polytope with twenty lattice points and $\delta(\Gamma_{32}) = 4$ for the one with fifteen, and among the four-polytopes with at most eleven lattice points the largest transition depth is 5.

8. **Free sums, exact families and growth with dimension**

The question of a lower bound increasing with the dimension has a constructive answer. The operation is the free sum in a *direct-sum lattice*. Its polarity and lattice-point behavior are standard; related enumerative identities are studied by Braun [2]. We give the arguments needed here, including the order isomorphism and the consequences for saturated descents. No classification or four-dimensional census is used in the proofs of this section's structural theorems.

Let M, N be lattices, and let $P \subset M_{\mathbb{R}}$ and $Q \subset N_{\mathbb{R}}$ be full-dimensional reflexive polytopes. Define

$$P \oplus Q = \operatorname{conv}\big((P \times \{0\}) \cup (\{0\} \times Q)\big) \subset (M \oplus N)_{\mathbb{R}}.$$

The direct-sum lattice is essential: we do not pass to an overlattice.

Lemma 25 (Lattice points of a free sum). *The free sum is reflexive and*

$$(P\oplus Q)^* = P^* \times Q^*, \qquad (P\oplus Q)\cap(M\oplus N) = ((P\cap M)\times\{0\})\cup(\{0\}\times(Q\cap N)).$$

In particular $|P \oplus Q| = |P| + |Q| - 1$.

Proof. A linear functional (u, v) is at least -1 on both factor polytopes exactly when $u \in P^*$ and $v \in Q^*$, proving the polar formula. The vertices of this product are pairs of lattice vertices, so the free sum is reflexive.

For a polytope P containing zero in its interior, write $g_P(x) = \inf\{t \ge 0 : x \in tP\}$ for its gauge. From the definition of the free sum,

$$(x, y) \in P \oplus Q \quad \Longleftrightarrow \quad g_P(x) + g_Q(y) \le 1. \tag{1}$$

Indeed a convex combination of points of the two factor polytopes gives nonnegative budgets a, b with $a + b = 1$, $x \in aP$, $y \in bQ$. Conversely, if the gauges sum to at most one, enlarge the two budgets to sum to one and use convexity and $0 \in P, Q$.

Zero is the only interior lattice point of a reflexive polytope. Therefore $g_P(x) \geq 1$ for $x \in M \setminus \{0\}$, and similarly for Q. A lattice point with two nonzero components would violate (1). The remaining lattice points are exactly those displayed. The origin belongs to both sets and is counted once. □

THEOREM 26 (Product of reflexive-subpolytope orders). *There is an order isomorphism*

$$R(P) \times R(Q) \longrightarrow R(P \oplus Q), \qquad (A, B) \longmapsto A \oplus B.$$

Every full-dimensional reflexive subpolytope of $P \oplus Q$ has a unique such factorization in the specified coordinate lattices.

Proof. Let $C \in R(P \oplus Q)$. Its vertices are lattice points, so Lemma 25 places each vertex on a coordinate factor. Let A and B be the convex hulls of its vertices in those factors. Both have zero in their interiors, relative to the full factor spaces. For if, for example, this failed for A, a nonzero supporting or separating linear functional on $M_{\mathbb{R}}$ would be nonnegative on A; extended by zero on $N_{\mathbb{R}}$, it would be nonnegative on C. This contradicts $0 \in \operatorname{int} C$. Consequently $C = A \oplus B$, where $A \subseteq P$ and $B \subseteq Q$ are full-dimensional lattice polytopes with zero inside.

Now $C^* = A^* \times B^*$ is a lattice polytope. A product has vertices exactly the pairs of vertices of its factors; projecting those vertices shows that A^* and B^* have lattice vertices. Hence A, B are reflexive. Conversely the free sum of any $A \in R(P)$ and $B \in R(Q)$ is a reflexive subpolytope by Lemma 25.

The intersections of $C = A \oplus B$ with the two coordinate subspaces are precisely A and B, since both contain zero. Thus the factors are unique. These intersections show that inclusion of free sums implies inclusion in each factor. The reverse implication follows directly from convex hulls, proving the order isomorphism. □

COROLLARY 27 (Covers change one factor). *The lower covers of $A \oplus B$ are exactly*

$$A' \oplus B \quad (A' \text{ a lower cover of } A), \qquad A \oplus B' \quad (B' \text{ a lower cover of } B).$$

Moreover $A \oplus B$ is r-minimal exactly when both factors are.

Proof. If both factors decreased, changing just one would supply a strict intermediate reflexive polytope. If one factor decreased by more than a cover, an intermediate in that factor would do the same. Conversely, when one factor is fixed and the other changes by a cover, the order isomorphism leaves no possible intermediate. The statement about r-minimal elements follows at once. □

Let $h(P)$ be the length of a longest saturated descent from P to an r-minimal polytope. Both $h(P)$ and $\delta(P)$ exist because $R(P)$ is finite.

THEOREM 28 (Exact additivity of descent lengths). *One has*

$$\delta(P \oplus Q) = \delta(P) + \delta(Q), \qquad h(P \oplus Q) = h(P) + h(Q).$$

Proof. Project a saturated descent to the two factor orders and omit repetitions. By Corollary 27 the projections are saturated descents, and their endpoints are r-minimal. Every step occurs in exactly one factor, so its length is the sum of the two projected lengths. This gives the lower bound for δ and the upper bound for h. Conversely, interleave two shortest descents, respectively two longest descents, keeping the other factor fixed at each step. Every step is a cover by the same corollary, and the final free sum is r-minimal. The two bounds are therefore attained. □

Remark 29 (Which chains are counted). We work with actual embedded subpolytopes in the fixed lattice, not with orbits of chains under automorphisms. A unimodular change of coordinates preserves the shortest and longest lengths, so the preceding equalities also compute the transition depth used for the classified polytopes. The counting statement below retains the embedded, labelled convention.

For $j \geq 0$, let $c_j(P)$ count all saturated descents of length j ending at an r-minimal polytope, and put

$$E_P(t) = \sum_{j \geq 0} c_j(P) \frac{t^j}{j!}.$$

This is a polynomial.

PROPOSITION 30 (Complete descent profile). *The profile satisfies*

$$c_n(P \oplus Q) = \sum_{a+b=n} \binom{n}{a} c_a(P) c_b(Q), \qquad E_{P \oplus Q}(t) = E_P(t) E_Q(t).$$

Proof. Choose a descent of length a in P, one of length b in Q, and the a positions of the first descent among the $a + b$ steps. The cover description gives a saturated product descent. Its projections recover both chosen descents and their interleaving uniquely. This is a bijection, which gives the convolution and then its exponential generating function. □

8.1. *A family with a direct proof and no finite census.* Consider

$$B = \operatorname{conv}\{a, b, c, d, e\},$$
$$a = (-1, -1), \quad b = (-1, 0), \quad c = (0, -1),$$
$$d = (0, 1), \quad e = (1, 0).$$

Its lattice points are these five vertices and zero. For example its primitive facet inequalities are

$$x + y \leq 1, \quad -x + y \leq 1, \quad x \geq -1, \quad y \geq -1, \quad x - y \leq 1.$$

Thus B is reflexive. Let $A = \operatorname{conv}(a, d, e)$, $H_b = \operatorname{conv}(a, b, d, e)$, $H_c = \operatorname{conv}(a, c, d, e)$, and $C = \operatorname{conv}(b, c, d, e)$.

PROPOSITION 31 (Five-element base order). *The embedded reflexive-subpolytope order of B consists of exactly A, H_b, H_c, C, B. The minimal elements are A, C. The three lower covers of B are H_b, H_c, C, and each H_b, H_c has the single lower cover A. Consequently*

$$\delta(B) = 1, \quad h(B) = 2, \quad c_1(B) = 1, \quad c_2(B) = 2, \qquad E_B(t) = t + t^2.$$

Proof. Any full-dimensional subpolytope with zero inside must contain e, the only available lattice point with positive first coordinate, and d, the only one with positive second coordinate. If it contains a, the choices are exactly A, H_b, H_c, B, according to whether b, c are included. If it does not contain a, it must contain both b, c to have negative coordinates in both directions; this gives C. The origin cannot be a vertex of a polytope in question, so no other choice is possible.

All five polytopes have zero in their interiors and are reflexive. This can be checked from their edges: A has inequalities $x - 2y \leq 1$, $x + y \leq 1$, $-2x + y \leq 1$; H_b has $x - 2y \leq 1$, $x + y \leq 1$, $-x + y \leq 1$, $-x \leq 1$; H_c has $-y \leq 1$, $x - y \leq 1$, $x + y \leq 1$, $-2x + y \leq 1$; and C has $\pm x \pm y \leq 1$ with all four choices of signs. All normals are primitive integral vectors. The inclusion relations are exactly those stated. In particular A and C contain no smaller reflexive full-dimensional polytope. Reading the five-element order proves the depth and profile formulas. □

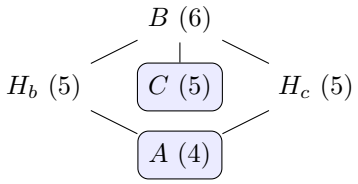


Figure 7. The complete embedded order for B. Parentheses give lattice-point counts; boxes mark r-minimal polygons. The left and right branches are different embedded chains even though an automorphism exchanges them.

THEOREM 32 (An explicit family of arbitrary depth). *For every $r \geq 1$, put $B_r = B^{\oplus r}$ in the lattice $\mathbb{Z}^{2r}$. Then*

$$|B_r| = 5r + 1, \quad \delta(B_r) = r, \quad h(B_r) = 2r, \qquad E_{B_r}(t) = t^r(1+t)^r.$$

More precisely, for $0 \leq j \leq r$ the number of descents of length $r+j$ is $(r+j)!\binom{r}{j}$, and there are none of other lengths. In dimension $2r+1$, $B_r \oplus [-1,1]$ has the same depths and profile.

Proof. Apply Lemma 25 and Theorem 28 repeatedly. The profile follows from Proposition 30 and $E_B(t) = t + t^2$; multiplying the coefficient of t^{r+j} by $(r+j)!$ gives the stated number. The interval $[-1,1]$ is the only one-dimensional reflexive polytope, is r-minimal, and has profile 1. A free sum with it changes neither the depths nor the profile. □

This theorem supplies a lower bound growing with dimension and an unbounded shortest–longest gap by a direct analytic construction. It uses no census, numerical hull computation, or imported certificate.

8.2. *A stronger small base checked independently.* The triangle

$$T = \operatorname{conv}\{(-1,-1), (2,-1), (-1,2)\}$$

has ten lattice points. Its depths $\delta(T) = 5$, $h(T) = 6$ also follow from the planar argument in Section 6. For the new profile calculation and an independent check of this base, we enumerate only the $2^9 = 512$ subsets of its nine nonzero lattice points. Two exact integer implementations are supplied. One takes monotone-chain convex hulls and closes them in $T \cap \mathbb{Z}^2$; the other uses all supporting directed pairs and no hull routine. They give the same 79 embedded reflexive subpolytopes. Their finite cover order has

$$c_5(T) = 150, \quad c_6(T) = 240, \quad c_j(T) = 0 \ (j \neq 5,6), \qquad E_T(t) = \frac{5}{4}t^5 + \frac{1}{3}t^6.$$

Appendix D specifies the finite calculation completely.

COROLLARY 33 (The triangle family). *For every $r \geq 1$, the explicit polytope $T_r = T^{\oplus r}$ satisfies*

$$\dim T_r = 2r, \quad |T_r| = 9r + 1, \quad \delta(T_r) = 5r, \quad h(T_r) = 6r,$$

$$E_{T_r}(t) = t^{5r}\left(\frac{5}{4} + \frac{t}{3}\right)^r.$$

Hence

$$c_{5r+j}(T_r) = (5r+j)!\binom{r}{j}\left(\frac{5}{4}\right)^{r-j}\left(\frac{1}{3}\right)^j \quad (0 \leq j \leq r).$$

The formula counts embedded chains, and its rational expression is an integer by its counting interpretation. A free sum with $[-1,1]$ gives the same conclusions in odd dimensions after increasing the dimension by one.

Proof. Use Theorem 28, Proposition 30, and the exact base profile just specified. No high-dimensional enumeration is needed. □

8.3. *The extremal depth as a function of dimension.*

THEOREM 34 (Superadditivity and an asymptotic limit). *The extremal depths are finite and satisfy*

$$D_{m+n} \geq D_m + D_n \quad (m, n \geq 1), \qquad D_d \geq \left\lfloor \frac{d}{2} \right\rfloor .$$

The independent triangle certificate strengthens the second inequality to

$$D_d \geq 5 \left\lfloor \frac{d}{2} \right\rfloor .$$

Moreover the extended limit exists:

$$\lim_{d\to\infty} \frac{D_d}{d} = \sup_{d\geq 1} \frac{D_d}{d} \ \in \ [5/2, \infty].$$

Using the earlier depth-22 result of Theorem 5 as a certificate-backed base further gives $D_{4r} \geq 22r$ and lower endpoint $11/2$ for this limit. The last strengthening imports precisely that earlier finite certificate, not the whole four-dimensional census.

Proof. Proposition 15 bounds all depths for $d \geq 4$. In dimensions two and three finiteness also follows by taking free sums with intervals to embed a given depth unchanged into dimension four. The one-dimensional depth is zero. Thus the nonempty set of possible integer depths in each dimension is bounded and has an attained maximum. Take polytopes attaining D_m, D_n and apply Theorem 28 to their free sum. This proves superadditivity. The lower bounds follow from Theorem 32 and Corollary 33.

For completeness, set $D_0 = 0$ for the following arithmetic argument. For fixed $k \geq 1$ write $d = qk + s$, $0 \leq s < k$. Superadditivity and nonnegativity give $D_d \geq qD_k + D_s \geq qD_k$. Thus $\liminf_d D_d/d \geq D_k/k$. Taking the supremum over k and observing that every D_d/d is at most that supremum proves the equality of extended limits, including the case of infinite supremum. The base inequality with $k = 2$ gives $5/2$. The same argument with the certificate-backed $D_4 \geq 22$ gives $11/2$ when that earlier result is used. □

The analytic family B_r alone settles the existence of a growing lower bound. The exact growth rate, whether the displayed limit is finite, and the optimal four-dimensional depth are not determined here. In particular $\delta(T \oplus T) = 10$ is an exact value for that particular four-polytope, not a replacement for the existing lower bound $D_4 \geq 22$.

9. Open questions

The original uniform-existence statement is proved. The exact four-dimensional maximum is still unknown: the certificate-backed range is $22 \leq D_4 \leq 346$ relative to the supplied census. The upper bound is not claimed sharp, and the heuristic of a few tens has not been established. Larger lower bounds require complete cover lists or another argument controlling every descent, rather than one displayed chain.

The question whether there is a lower bound growing with dimension now has an affirmative answer by Theorem 32. Theorem 34 gives superadditivity and an extended limit for D_d/d, but does not decide whether the limit is finite or give the optimal rate. The upper bound remains doubly exponential.

The shortest–longest gap is unbounded as dimension varies: B_r and T_r have gap r. Its largest value in each fixed dimension is not determined by that construction. In dimension three the earlier computation gives a largest gap of 20, for a 35-point simplex of depth 8 and longest descent 28 (Fig. 4).

Whether a polytope attaining D_4 must be r-maximal remains open. The two-dimensional extremizer is the ten-point triangle; both three-dimensional extremizers are r-maximal. The known depth-22 polytope P_1 lies inside Δ_{680} and hence is not r-maximal. The free-sum theorem describes exact families, without classifying all extremizers. The unresolved higher-dimensional nonreflexive-cut behavior in Section 5 is likewise not settled by these families.

Data availability

The original code and data are archived in release 1.2.0 of the repository, https://github.com/creelie/seed/releases/tag/v1.2.0 [22]. The archive contains the joint distribution of point counts over the whole list, the pair bounds, the cover relation in dimensions two and three, the descent of Δ_{680} with its certificates, the certificate of Lemma 22 for P_1 and P_2, the checks of the cuts formed by Corollary 19 in dimension three and at the certificate polytopes, and independent checks in C, C++, Python, Julia, Macaulay2, shell and Lean 4. The Kreuzer–Skarke list was read from its public copy [12]. The search for long descents was organised on the Anaxia engine, release 3.0.5, and rechecked on release 3.1.1; the engine is archived in release 3.2.1 [20] and described in [21], and its record of strategies, failed checks and the changes they forced is part of the archive. The original computations and text used Claude (Anthropic). For this review revision, ChatGPT (OpenAI) assisted with the free-sum proofs, exact-integer checks and manuscript preparation. The new `anc/` directory contains the two base certificates, their standard-library Python verifier, the interval verifier and its data, and the current verification report. These additions accompany release v1.3.0 of https://github.com/creelie/seed/releases/tag/v1.3.

0; its version DOI is recorded in the archive receipt. The complete 5,592-node cover reconstruction from the preceding paper was not rerun in this revision. Its earlier certificate remains an explicitly imported finite base for the stronger $22r$ lower bound. The independent pentagon and triangle families do not require it. The core geometric order isomorphism and depth-additivity results now have Lean proofs, with the precise scope given in Appendix E. The whole manuscript is not formalized.

Statements and Declarations

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Competing interests. The author declares no competing interests.

Appendix A. The reflexive polygons

Table 2 lists the sixteen reflexive polygons by their vertices, with the number of lattice points, the number of covers, the transition depth δ and the length h of a longest descent. For each polygon with at least six points the last column gives the vertices of an r-minimal polygon with five lattice points inside it, which is what the proof of Theorem 4 needs. Each entry can be checked by hand.

Appendix B. The descent from the largest polytope

Table 3 lists the descent of Section 7: for each step i the number of lattice points of Γ_i and of its dual and the number of vertices of Γ_i. Steps 1 to 10 were certified by the climbing lemma and steps 11 to 38 by listing all covers. The lattice points of every Γ_i are in the archive [22].

Appendix C. How the computations were checked

Every computational statement in the paper was obtained by one method and confirmed by a second, written independently, where that was practical.
Point counts. The joint distribution of $|\Delta|$ and $|\Delta^*|$ was read from all 473,800,776 entries of the list [12]. The total, the extreme values 6 and 680, and the single polytope at $(680, 26)$ agree with Kreuzer and Skarke [6]. The pair bound was computed by the recursion of Section 4 and again by a direct quadratic search over all pairs, with the same result; a separate program checks the bounds 654 and 348 and the counts of Table 1.
Covers. In dimension two the sixteen polygons and their covers were computed from scratch and agree with Table 2. In dimension three the cover relation obtained by removing lattice points and the one obtained from Corollary 19 agree on every polytope. Unimodular equivalence was decided with the normal forms of PALP [7].

The simplex Δ_{680}. Its 680 lattice points, the 26 points of its dual and the volumes 3528 and 84 were checked by four separate programs, and the arithmetic of the weights by a proof assistant. Each step of the descent in Table 3 was certified as a cover, and the last polytope was checked to have no reflexive proper subpolytope.

Exact depths. The values $\delta(\Gamma_{27}) = 9$ and $\delta(\Gamma_{32}) = 4$ come from breadth-first searches through complete lists of covers, in which unimodularly equivalent polytopes are identified. The search stops at the first level that contains a polytope without covers, so it gives the least length of a descent.

The certificate for P_1 and P_2. The archive stores the 5,592 polytopes of the certificate by their lattice points and PALP normal forms, the list of covers of each of the 5,250 polytopes with positive bound (23,646 covers in all) and of one polytope without covers, the bound of every polytope, and the two descents of length 22. A separate program checks, without PALP, that every stored point set is the set of lattice points of a reflexive polytope, that the roots are P_1 and P_2, that every cover has fewer lattice points than the polytope it covers, that the bounds satisfy the hypothesis of Lemma 22 and equal 22 at the roots, that both descents end at a polytope without covers, and that the matrix G of Proposition 24 is unimodular and carries the lattice points of P_1 onto those of Γ_{10}. With PALP it recomputes every normal form and every list of covers by Corollary 19, and all 5,251 recomputed lists agree with the stored ones. The descent of length 22 from Γ_9 is stored as a chain of subpolytopes of Γ_9, and each of its steps is checked to be a cover in the same way. Lemma 22, the point counts along the descents and the arithmetic of G are also theorems of the Lean file.

Every dimension. The step from Blichfeldt's inequality to the bound of Proposition 15 is a theorem of the Lean file for every d and every V: a chain of point counts that strictly decrease, stay at least $d+2$ and start at most at $V+d$ has at most $V-2$ steps. A short program compares the bound with the posets of dimensions two and three, whose longest chains of covers have 6 and 28 steps, and with the point counts of the list in dimension four.

Appendix D. The exact planar base certificate

We specify the independent finite calculation behind the stronger base. The nine boundary points of T, in the order used by both checkers, are

$$(-1,-1),(-1,0),(-1,1),(-1,2),(0,-1),(0,1),(1,-1),(1,0),(2,-1).$$

A mask $m = \sum_{i=0}^{8} \varepsilon_i 2^i$ selects a subset of these points. All 512 masks are examined. A polygon is retained only if zero is strictly interior and its point set is closed in $T \cap \mathbb{Z}^2$. Reflexivity is checked directly from primitive edge distances, not from a floating-point hull calculation.

D.1. *Integer hull and supporting-pair algorithms.* The first implementation orders selected points lexicographically and uses the monotone-chain algorithm: remove the middle point of any non-left turn until both boundary chains are strictly convex. Every test is the sign of

$$[a, b, p] = (b_1 - a_1)(p_2 - a_2) - (b_2 - a_2)(p_1 - a_1).$$

It then adds every point of $T \cap \mathbb{Z}^2$ on or to the left of every oriented hull edge. The resulting point mask is the canonical closed representative. Every possible reflexive subpolygon occurs: its vertices are a subset of the nine listed points and are included among the masks.

The second implementation does not call that hull routine. For each selected set with a noncollinear triple it lists every ordered pair (a, b) with $a \neq b$ such that $[a, b, p] \geq 0$ for all selected p. These are exactly the oriented supporting lines determined by pairs of selected points. For all such pairs it requires

$$[a, b, 0] = \gcd(|b_1 - a_1|, |b_2 - a_2|).$$

The right side is positive, so this condition puts zero strictly inside and makes the primitive lattice distance of every edge equal to one. For each unselected boundary point it also requires a supporting pair with $[a, b, p] < 0$; hence no point of the hull has been omitted. This tests closedness without any convex-hull implementation. The two algorithms return exactly the same 79 masks.

D.2. *Cover lists, depths and full profiles.* For two closed masks, inclusion is the bitwise test $a \,\&\, b = a$. The proper submask a is a cover below b exactly when no third retained mask c satisfies $a \subsetneq c \subsetneq b$. The finite list therefore determines every cover, not just a proposed descent.

Compute profiles in increasing numeric mask order, which is topological for proper inclusion. A minimal mask has polynomial 1. For a nonminimal mask b the *ordinary* length-count polynomial is

$$C_b(z) = z \sum_{a \text{ lower cover of } b} C_a(z).$$

No multiplicity is discarded: covers here are actual embedded point sets. The full mask 511 has $C_{511}(z) = 150z^5 + 240z^6$. The 79 depths are distributed as follows:

depth	0	1	2	3	4	5
number	14	33	19	9	3	1

The fourteen minimal masks are

$$78, 88, 98, 113, 116, 135, 148,$$
$$161, 178, 184, 198, 262, 340, 418.$$

Each has four or five lattice points. Every cover in this embedded order loses exactly one point. Thus the longest possible route from the ten-point root to a minimal polygon has six steps, while a shortest route has five. For example the following vertex sets give a five-step descent, with point counts $10, 9, 8, 7, 6, 5$:

$$\{(-1,-1),(2,-1),(-1,2)\},$$
$$\{(-1,-1),(1,-1),(1,0),(-1,2)\},$$
$$\{(-1,-1),(1,-1),(0,1),(-1,2)\},$$
$$\{(-1,-1),(1,-1),(-1,2)\},$$
$$\{(-1,0),(0,-1),(1,-1),(-1,2)\},$$
$$\{(-1,0),(1,-1),(-1,2)\}.$$

The lower bound on all descents comes from the complete cover list, not from displaying this one path.

The file `triangle-certificate.json` contains all retained masks, vertices, complete point sets, covers, depths, heights and profiles. The program `verify_free_sums.py` regenerates it using only the Python standard library. It independently enumerates the 32 subsets for the pentagon base and checks its five-element order and profile. As further controls it enumerates the lattice points of $T \oplus T$ from its gauge inequalities, obtaining 19 points, and calculates the shortest and longest paths on the $79^2 = 6241$-element product cover graph, obtaining 10 and 12. These are controls for the written geometric proof; no new complete Lean formalization is claimed.

Appendix E. Scope of the Lean verification

The release includes a pinned Lean project under `formal/`, with Lean 4.34.0-rc2 and Mathlib commit `85e3a25e006c35636f0e53b0e9296caca2685bc0`. Its thirteen positive modules contain eighty-one named source theorems. The axiom audit covers 255 theorem declarations including generated auxiliary declarations; this is not a count of independent results. All target dependencies are among the standard foundational axioms `propext`, `Classical.choice` and `Quot.sound`. There are no admitted target proofs, custom axioms or uses of `native_decide`. Lean's kernel replay tool rechecks the new module declarations relative to their imports. The logs, source hashes and exact coverage are part of the release.

The geometric definitions use actual real convex hulls, additive lattices, topological interiors and bilinear polar pairings. Reflexivity uses the IP convention: zero is the unique interior lattice point and both the polytope and its polar are finite lattice hulls. The difference between the upper-bound polar convention used by the code and the lower-bound convention of this paper is reconciled by a proved negation equivalence. The product of subpolytope orders is proved

from these definitions, rather than assumed as an axiom. The formal project also proves the exact decomposition of saturated descent lengths and shortest and longest length addition, finite-order termination given a complete finite ambient lattice-point list, the equivalence of terminal elements and r-minimal elements, the interval lemma and the superadditive-sequence limit lemma.

The stored depth-22 graph is checked separately. The kernel checks all its 5,592 nodes and 23,646 recorded edges, and verifies shortest graph length 22 for both recorded roots. A negative control that raises the first root's lower bound to 23 is rejected as false. Null successor lists at zero-bound frontier nodes are treated as empty in this finite graph; this does not assert geometric r-minimality of those frontier polytopes. The lower certificate uses the positive-bound nodes. Both displayed paths end at a node with an actually empty list.

The graph check is not an end-to-end formal verification that every recorded list is the complete list of geometric covers. The concrete pentagon order identification, triangle enumeration and profile counts, the four-dimensional census and twelve interval geometries, and the published volume/classification inputs are not all formalized here. In particular the range $22 \leq D_4 \leq 346$ is not claimed as a single end-to-end Lean theorem about all reflexive four-polytopes, and the exact value of D_4 is not determined. The full coverage document is `formal/FORMAL-COVERAGE.md` in the release.

A limitation of free sums in dimension four. Using $D_1 = 0$, $D_2 = 5$ and $D_3 = 11$, Theorem 28 shows that a nontrivial four-dimensional free sum has depth at most 11: its factor dimensions are $(1, 3)$, $(2, 2)$ or $(3, 1)$ and the corresponding bounds are 11, 10 and 11. Therefore a depth-22 four-polytope cannot be a nontrivial free sum in its lattice, even after a unimodular change of coordinates. Passing to an overlattice does not preserve the argument: the free sum of $[-1, 1]$ with itself contains $(1/2, 1/2)$ in the lattice generated by $(1, 0)$, $(0, 1)$ and $(1/2, 1/2)$. This is a mixed lattice point on the diamond and violates the axis-support conclusion of Lemma 25 when the direct-sum lattice hypothesis is dropped. The formal project checks this explicit counterexample and the low-dimensional numerical bound, without claiming a new proof of the low-dimensional maxima.

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